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Enhancing Hallucination Detection in Large Language Models by Internal-External Feature Fusion

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***Abstract*—Detecting non-factual contents in Large Language Model (LLM) generations is the first step for mitigating the hallucination in LLMs. Current probing-based methods rely only on human-annotated labels demonstrating and suffer from limited transferability to out-of-distribution contents, while the consistency-checking based methods fail to correctly detect hallucinations when the LLM consistently produces incorrect outputs. This paper proposes ESIF, which leverages external semantic distribution features to reduce dependency of probing-based methods on data distribution, thereby enhancing the transferability of LLM. In addition, ESIF fuses external and internal features to train the classifier, which improves the accuracy of hallucination detection. These two types of features work in a complementary manner: external features help reduce the reliance on data distribution by capturing the overall semantic trends, while internal features enhance this framework by refining the detection of edge cases in which rely on external features solely may fall short. Experiment results demonstrate that the proposed ESIF outperforms the existing state-of-the-art methods in hallucination detection and achieves excellent transferability.**

***Index Terms*—LLM, Hallucination Detection, Fact Checking.**

I. Introduction

The rapidly developing Large Language Models (LLMs) in recent years such as GPT-4 [1], LLama [2], GLM [3], have demonstrated remarkable performance in a variety of natural language processing (NLP) tasks. However, LLMs suffer from the critical hallucination problem that concerns their authenticity. Hallucination in LLMs describes the phenomenon that the generated content of an LLM is nonsensical or unfaithful to the provided source content [1], [4], [5]. This hallucination problem seriously harms the effectiveness of LLMs in practical applications, and could even lead to serious consequences in some cases, such as the healthcare QA scenarios [6], [7]. As hallucination in LLMs seriously undermines their reliability in real-world NLP tasks, many efforts have been devoted to hallucination detection and mitigation in LLMs.

Previous research has revealed that the hidden layer activations of the answers generated by LLMs can be used as a basis to detect whether they are hallucinated or not [8]. These approaches, commonly referred to as probing-based methods, analyze the internal representations and activations within the LLMs to infer the reliability of the generated contents. However, they often have limited transferability to out-of-distribution data, as the internal states are typically tuned for the specific distribution of the training data. This limitation restricts their effectiveness in the scenarios involving new or unseen data distributions. For instance, reliance on token-level features derived from LLMs often results in suboptimal performance when applied to new datasets with differing semantic structures [5]. For instance, suboptimal performance was observed when token-level features derived from LLMs were applied to new datasets with different semantic structures [5], [9], [10]. Similarly, when the question is outside the training domain, the effectiveness of the intrinsic uncertainty metrics designed for in-distribution hallucination detection is degraded [4]. These observations highlight the need for new approaches that can generalize better to diverse data distributions without overfitting to specific training scenarios.

To overcome the transferability limitations of probing-based methods, recent studies aim at detecting the hallucinations from the output of LLMs. By comparing the output with an external reliable real corpus [11], [12] or taking other self-comparison approaches [13], [14], the presence of hallucinations in the output of LLMs can be discerned to some extent. The former relies on the scope of the external knowledge base or the LLMs' own knowledge. Thus, there is still a high probability that the LLMs output non-factual content when the input question is out of this scope [15]. The latter is not as effective as the probing-based methods due to the lack of a training process. For example, by sampling multiple responses from the LLM, SelfChekGPT can measure the consistency of information between different responses and determine whether the utterance is factual or illusory [16]. However, it may make wrong judgements when LLMs output wrong yet consistent contents.

To tackle these limitations,we propose **ESIF**, a method for hallucination detection by employing both external semantic distribution and internal features. The main idea of ESIF is to fuse internal and external features to train the classifier for hallucination detection. We highlight ESIF with the following two advantages:

Transferability. Compared to the traditional probing-based methods, ESIF reduces the dependence of LLMs on data

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distribution by introducing external semantic distribution state as a new feature. This implies that ESIF only needs to be trained on a single model and dataset, while maintaining strong performance on other models and datasets without the need for retraining.

Efficiency. Compared to previous approaches, ESIF increases the accuracy of hallucination detection by integrating external semantic distribution features with internal model features. Additionally, to the best of our knowledge, this is the first attempt to fuse internal and external features for improving hallucination detection in LLMs.

By experimenting on different open datasets and models, our results reveal several key findings: (1) ESIF outperforms the existing methods on all datasets; (2) The results on different datasets reflect the excellent transferability of ESIF; (3) The introduction of the attention mechanism for both internal and external features in our ESIF improves its accuracy of hallucination detection.

II. Realated Work

In this section, we provide a review of hallucination detection methods proposed in previous works.

A. Probing-based Methods

One category of hallucination detection methods are the probing-based methods, which leverage the internal states of LLMs to assess the reliability of their outputs. These methods analyze the intrinsic uncertainty encoded within the hidden layers of the LLMs in the process of response generation. For example, Azaria et al. [8] demonstrated that the implicit activation of the answers generated by LLMs can serve as a basis for detecting hallucinations. Su et al. [17] proposed an unsupervised training framework to utilize the internal states of LLMs for real-time hallucination detection. By focusing on the internal features, these approaches aim to detect hallucinations without relying heavily on external data. However, the effectiveness of the probing-based methods can vary depending on the layers of the model being analyzed and the specific task. Moreover, their generalizability to diverse contexts still remains a challenge. In practice, selecting which layers to probe and determining how to integrate these signals with other strategies is non-trivial. Ongoing research seeks to refine these methods to make them more adaptable and stable across various model architectures and application scenarios so as to gradually expanding their applications in real-world settings.

B. Consistency-checking Method

Another category of hallucination detection methods are the consistency-checking methods, which determine the presence of hallucinations by evaluating the consistency of LLM outputs with external reliable corpora or through self-comparison techniques [11], [12]. For instance, SelfCheckGPT [16] generates multiple random samples using the same query and evaluates the consistency among these responses. SelfCheckGPT employs two approaches: one based on BERTScore [18], and the other using a multiple-choice question answering and generation method (MQAG) [19]. Kadavath et al. [20] evaluated model self-consistency through model confidence in generated contents. This involves prompting the model to internally evaluate the accuracy of its prior predictions and estimating the likelihood that its responses are correct. However, consistency-checking based methods often rely heavily on the availability and scope of the external knowledge base or the LLMs' own knowledge, making them less effective in the inputs that fall out of these scopes [15].

III. Problem Formulation

In this study, we address the challenges of hallucination detection in LLMs by combining internal and external features to assess the factuality of model-generated contents. Hallucination in LLMs refers to the instances where the generated text deviates from factual correctness or logical coherence. Our approach leverages the internal features extracted from the hidden states of the model during response generation and the external features extracted from the semantic distribution of multiple responses to the same question. Given a natural language question, our objective is to determine whether the primary response is reliable by integrating these two complementary feature sets.

Detection Goals. The problem is formulated as a binary classification task to identify whether the primary response generated by an LLM is factual or hallucinatory. For a given question, the LLM produces a response r_{primary}, and its internal hidden states of layer h_s during generation are recorded. These internal features f_{internal} provide insights into the model's confidence and semantic alignment.

Simultaneously, we analyze the semantic distribution of the primary response in comparison with additional responses:

$$R_{\text{all}} = \{r_1, r_2, \ldots, r_n\} \tag{1}$$

generated through diverse settings. The statistical properties of this distribution, such as mean and variance, serve as external features f_{external} that indicate the reliability of the response.

Finally, the hallucination detection framework is constructed by combining the internal and external features. Let the combined feature vector be x:

$$x = a_1 f_{\text{internal}} + a_2 f_{\text{external}} \tag{2}$$

where a_1 and a_2 are the weights of the attention that balance the contribution of internal and external features.

The final prediction is made by applying a binary classifier to the feature vector x:

$$\hat{y} = \text{classifier}(x) \tag{3}$$

where $\hat{y} \in \{0, 1\}$ represents the predicted label. Specifically, 0 indicates a factual response and 1 indicates a hallucinatory response.

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Language Model Probing. The internal states of an LLM, particularly the hidden layer states, encode valuable information about the model's semantic and syntactic processing. These states provide a window into the model's "belief" in the factuality of its outputs. When a natural language question is posed to the LLM, it generates a response and produces hidden states at each generation step simultaneously. These states vary across layers with earlier layers capturing semantic features closely tied to the input and later layers influenced by the generated output. By analyzing these hidden states, meaningful internal features that reflect the model's underlying reasoning during response generation can be extracted, which are crucial for understanding the reliability of the generated text.

Distribution Statistics. Semantic distribution analysis provides an external perspective on the reliability of LLM-generated responses. For a given question, the LLM produces multiple responses, including the primary response and the additional responses generated by the techniques like temperature tuning and varied prompts to enhance the diversity. The semantic similarities between the primary response and the additional responses can be calculated using cosine similarity. Reliable responses tend to exhibit a tightly clustered distribution of similarity scores with high mean and low variance, reflecting a strong semantic consistency. By contrast, hallucinated responses usually show either a dispersed distribution or low mean similarity, indicating a lack of coherence.

By analyzing external semantic distribution features and combining them with internal features to train a classifier, our approach provides a robust and interpretable framework for detecting the hallucinations in LLMs. This foundation allows for effective classification of reliable and unreliable outputs, paving the way for more trustworthy language generation systems.

IV. Methodology

A. Overview of ESIF

The implementation of the ESIF framework comprises three primary stages: (1) In the internal feature extraction stage, we input natural language questions into a LLM to generate answers, and obtain the states of the hidden layers during the generation of answers; (2) In the external semantic distribution analysis stage, we characterize the semantic distribution of multiple answers to the same question; (3) In the classifier training stage, we train the classifier jointly by the internal and external features obtained in the first two stages. Figure 1 illustrates the overall process.

B. Stage 1: Internal Representation Extraction

The first stage of ESIF leverages internal states of large language models (LLMs) to capture latent information relevant to hallucination detection.

Let a user query be denoted by q, and let $\mathcal{M}$ represent a frozen LLM (e.g., LLama2-7b). The model generates a primary response $r_0 = \mathcal{M}(q)$ and simultaneously produces hidden states at each transformer layer $l \in \{1, \ldots, L\}$, denoted as:

$$\mathbf{H}_l(q) \in \mathbb{R}^{T \times d}, \tag{4}$$

where T is the sequence length and d is the dimension of the hidden layer. These hidden states encode the model's internal confidence or belief about the truthfulness of its outputs.

Identifying the optimal layer l^* for representation extraction is non-trivial. Lower layers tend to encode syntactic and semantic patterns from q, while higher layers are influenced by the generation process. Following Azaria et al. [8], we use the 16th layer ($l^* = 16$) in LLama2-7b. The hidden state is aggregated via mean pooling into a vector $\mathbf{z}_{\text{int}} \in \mathbb{R}^d$:

$$\mathbf{z}_{\text{int}} = \text{Aggregate}\left(\mathbf{H}_{l^*}(q)\right) \tag{5}$$

C. Stage 2: External Semantic Distribution Analysis

The second stage of ESIF focuses on analyzing the semantic distribution of multiple responses generated to the same question. This is achieved in two steps:

Multi-response generation. For given q, we generate k diverse responses $\{r_1, \ldots, r_k\}$ using bootstrapping techniques, including: (1) **Temperature Tuning** : Tuning the parameter temperature τ in the LLM is an effective way to control the diversity of the output. Temperature indicates the stochastic nature of the model's predictions: the lower the temperature (e.g., near 0), the more deterministic the model is, favouring words with higher probabilities and thus producing more focused and repetitive responses. By contrast, the higher the temperature, the greater the stochasticity, allowing the model to choose from a wider range of possible words, thus producing more varied and creative results. By peoperly tuning the temperature, we can make the output of the LLM more diverse. (2) **Prompt Design**. We develop multiple prompt templates $\{p_1, \ldots, p_k\}$ to guide the LLM in generating rich and diverse responses. These templates introduce subtle variations in the phrasing and structure of questions, encouraging the model to explore different semantic directions in its answers.We define the full set of responses as:

$$\mathcal{R}(q) = \{r_0\} \cup \{r_i = \mathcal{M}_{\tau_i}(q_i)\}_{i=1}^k \tag{6}$$

Semantic Distribution Analysis. Each response r_i is encoded using a sentence encoder f_{enc} to obtain $\mathbf{v}_i = f_{\text{enc}}(r_i) \in \mathbb{R}^{d'}$. Cosine similarity between the primary response and each alternative is calculated as:

$$s_i = \cos(\mathbf{v}_0, \mathbf{v}_i) = \frac{\mathbf{v}_0 \cdot \mathbf{v}_i}{\|\mathbf{v}_0\| \|\mathbf{v}_i\|}, \quad i = 1, \ldots, k. \tag{7}$$

We then compute the mean and variance of these similarity scores:

$$\mu_s = \frac{1}{k} \sum_{i=1}^k s_i, \quad \sigma_s^2 = \frac{1}{k} \sum_{i=1}^k (s_i - \mu_s)^2. \tag{8}$$

These form the external feature vector:

$$\mathbf{z}_{\text{ext}} = \begin{bmatrix} \mu_s \\ \sigma_s^2 \end{bmatrix} \in \mathbb{R}^2. \tag{9}$$

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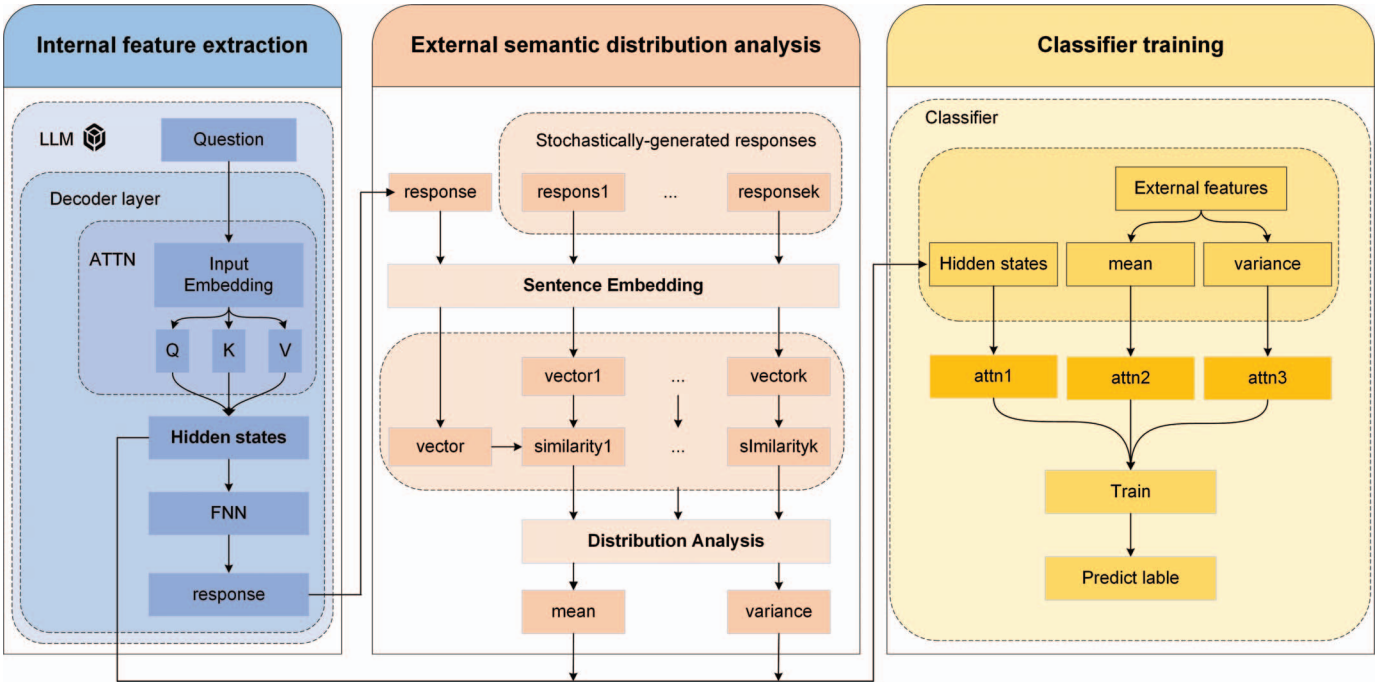


Fig. 1. The overall architecture of ESIF

While semantic distribution provides valuable insights, it has limitations. For instance, if the LLM consistently generates incorrect responses for a question, the distribution may still appear tightly clustered at a high mean. To address this, the internal hidden states from the first stage are combined with external distribution features to enhance the reliability of hallucination detection.

D. Stage 3: Classifier Training

In the third stage, we concatenate internal and external features into a unified representation:

$$\mathbf{z} = [\mathbf{z}_{\text{int}}; \mathbf{z}_{\text{ext}}] \in \mathbb{R}^{d+2}. \tag{10}$$

This feature vector is used to train a binary classifier $\mathcal{C}$ for hallucination detection:

$$\hat{y} = \mathcal{C}(\mathbf{z}) \in \{0, 1\}. \tag{11}$$

Specifically, we implement $\mathcal{C}$ as a feed-forward neural network with hidden sizes of 256, 128, and 64, ReLU activations, and a sigmoid output layer. To dynamically weigh internal and external contributions, we apply an attention mechanism:

$$\mathbf{z}' = \alpha \cdot \mathbf{z}_{\text{int}} + (1 - \alpha) \cdot \mathbf{z}_{\text{ext}}, \quad \alpha \in [0, 1], \tag{12}$$

where weight α is learned during training. The model is optimized with the Adam optimizer, without extensive hyperparameter tuning, highlighting the robustness of our framework.

E. Discussion

To validate the feasibility of the external semantic distribution features, we conduct an empirical analysis of the similarity mean and variance distributions for both positive and negative samples. From the training and test datasets, we randomly sample 2,500 positive (reliable responses) and 2,500 negative examples (hallucinatory responses), and plot the distributions of similarity mean and variance.

It can be seen that in Figure 2, positive and negative samples exhibit distinct patterns. Positive samples exhibit higher similarity mean and lower variance, indicating tightly clustered semantic consistency, whereas, negative samples display the opposite pattern with lower mean similarity and higher variance, suggesting a lack of coherence.

These findings confirm the validity of exploring external semantic distribution for hallucination detection. Moreover, by integrating external features with internal hidden layer states, the classifier can overcome the limitations of utilizing external features only for hallucination detection. This fusion further strengthens the classifier's ability to accurately detect hallucinations. Future work will refine the feature extraction and classification processes to further improve the detection performance across diverse datasets.

V. Experiments and Results

A. Dataset

We employ the training set used for PINOSE [21], which is specifically tailored for question-answering tasks to train our ESIF. Particularly, this dataset is ideal for developing models that are capable of handling a broad spectrum of questions across various knowledge domains. Its diversity not

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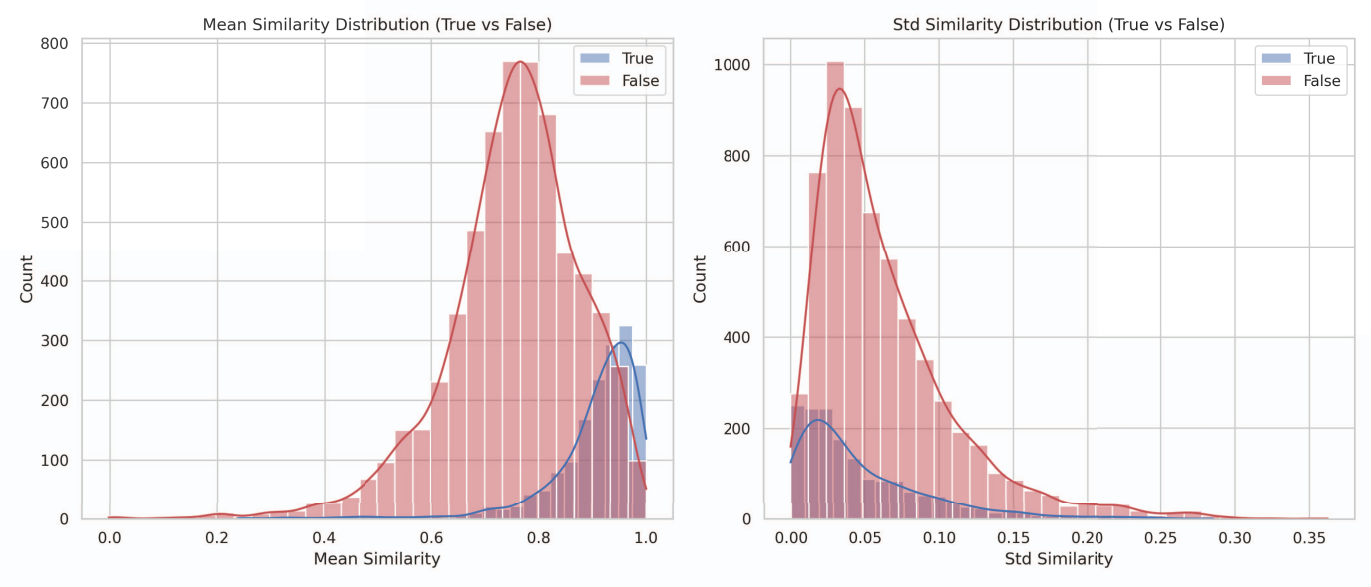


Fig. 2. Distribution of similarity mean and similarity variance for positive and negative samples

TABLE I
STATISTICS OF DATASETS.

	True-False	NQ	TriviaQA	WebQ
Train	5000	N/A	N/A	N/A
Test	1000	1000	1000	1000
True Ratio	40.5	46.6	49.5	58.3

only provides a robust foundation for training models like ESIF to identify inconsistencies in responses, but also ensures that the model is exposed to a wide range of response patterns.

For evaluation, we test ESIF on three widely recognized datasets: NaturalQuestions [22], TriviaQA [23], and WebQ [24]. These datasets are well-established in the field and known for their comprehensive coverage of question-answering challenges. Specifically, we randomly select 1000 pieces of data from each dataset separately to construct the test set. The statistical details of the processed datasets are presented in Table I.

B. Baselines

We compare ESIF with the representative probing-based and consistency-checking based methods. For a more comprehensive comparison, we also implement confidence-based methods as baselines:

- **Probing-based method: SAMPLA** [8] is a hallucination detection method, trained using the activation values derived from the hidden layers of LLMs. **PINOSE** [21] introduces an efficient method for detecting non-factual contents in LLMs by training a probing model on offline self-consistency checks, bypassing the need for human annotations and reducing the computational demands. By analyzing the internal states before response generation, PINOSE effectively detects factual inaccuracies and enhances the model reliability across varied data distributions. **RepE** [25] uses internal representations of language models to probe their decision-making process. By systematically examining these representations across different layers, RepE aims to identify the patterns and correlations that can explain the model's behavior. This method focuses on extracting useful information from both the hidden states and token embeddings, offering transparency into how LLMs generate outputs.
- **Consistency-checking based method: SelfCheck-GPT(SCG)** [16] is designed based on the principle that if an LLM has the knowledge of a given concept, sampled responses are likely to be similar to each other and contain consistent facts. SCG employs four distinct techniques: SCG_BERTScore, SCG_QA, SCG_NLI, and SCG_n-gram, all of which evaluate the consistency of the responses from the LLM.
- **Confidence-based method: Perplexity-AVE** and **Perplexity-Max** [20], [25] measure the mean and the highest token-level probabilities of statements produced by the assessed LLM in each test set. **It-is-True** [8] compares the probability values between the statements "It is true that q concatenated with r" and "It is false that q concatenated with r."

C. Metrics

We follow the conventions in factuality detection and employ the area under the receiver operating characteristic curve (AUC) and accuracy (ACC) as evaluation metrics.

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TABLE II
OVERALL PERFORMANCE ACROSS THREE TEST DATASETS

Method	NQ		TriviaQ		WebQ	
	AUC	ACC	AUC	ACC	AUC	ACC
RepE	62.6	60.8	67.5	64.3	65.5	63.5
PINOSE	79.4	73.1	82.9	69.6	82.8	73.2
SAMPLA	69.7	67.8	69.3	64.5	75.6	69.5
SCGPT-PT	72.3	70.3	76.4	69.9	72.5	72.5
SCGPT-NLI	77.4	71.0	76.9	72.1	77.4	72.3
PPL-AVE	61.8	59.2	61.7	58.5	63.7	63.3
PPL-MAX	54.2	55.0	57.6	57.0	56.4	60.0
It-is-True	63.0	59.5	61.7	59.1	75.6	71.0
ESIF w/o attn	78.7	71.3	**83.7**	68.5	82.5	72.4
ESIF w/ attn	**81.4**	**74.4**	82.5	**72.4**	**83.1**	**76.9**

D. Implementation Details

To implement ESIF, the LLM is invoked several times during the building process, including the internal feature extraction stage, the external semantic distribution analysis stage and the final non-factual detection stage. The same LLM is used at all stages to ensure that the model's understanding and feature extraction remain consistent across the entire process, reducing potential discrepancies or biases that could arise from using different models. Note that consistency helps maintain accurate non-factual detection, as it ensures that the same semantic space is used throughout.

E. Main Results

Table II presents the performance comparison between our proposed method and several factuality detection baselines across three datasets (NQ, TriviaQ, and WebQ) in terms of AUC and ACC, evaluating.

ESIF *w/ attention* consistently achieves the highest AUC and ACC values across all datasets, with an AUC of 81.4 and ACC of 74.4 on NQ, and an outstanding AUC of 83.1 and ACC of 76.9 on WebQ. These results not only surpass those of the baselines, RepE and SAMPLA, but also outperform those of the advanced models such as PINOSE and SCGPT variants, emphasizing the accuracy and robustness of our approach.

External features improve transferability: It is shown in Table II that although the traditional probing-based methods such as Sampla also employ supervised training, they perform worse than our proposed ESIF on the three test sets. This is due to the formal gap between the dataset employed for training and the dataset employed for testing, i.e., different data distributions. To tackle this, ESIF introduces external semantic distribution features that are independent of the data distribution. Specifically, ESIF obtains semantic features by sampling the same problem several times, and then uses these external features along with the internal features as the inputs for training, which improves its robustness in achieving good performance across different datasets with different distributions. PINOSE also adopts a consistency-based approach, but it is only used to lable the data and not involved in the classifier training, making it less effective than ESIF.

Internal features improve accuracy: SelfcheckGPT uses a consistency-based approach, and achieves better results than the probing-based approach although there supervised training is adopted. This proves the effectiveness of consistency checking in checking facticity. But consistency-checking based methods may fail under certain conditions, particularly when the LLM consistently makes incorrect judgments. Our ESIF also explores external semantic distribution for consistency checking, yet it incorporates internal features to mitigate this issue. The internal features provide additional contextual information and enables the model to make more accurate judgments, even when external consistency checks exhibit biases. This effectively improves the accuracy of the system.

Impact of attention mechanism: The integration of attention mechanism in "ESIF *w/ attention*" is particularly impactful, as evidenced by its superior performance over the base version of our model and other comparative models. By allowing the model to prioritize the relevant information within the responses, the adaption of attention mechanism enhances factuality detection, leading to more reliable and contextually-aware predictions.

F. Ablation Studies on Model

In the ESIF implementation, we default to using the same LLM for both internal and external feature extraction. The starting point of this design is the belief that using the same LLM for both tasks maintains the consistency in data representation and thus contributes to the consistency of the detection process. However, to further explore the possible effects of using different LLMs in the two sessions, we conducted an ablation experiment. Specifically, we use different combinations of LLama2-7B and LLama2-13B for the internal and external feature extraction tasks, respectively, in our experiment.

The results of the experiments are shown in Table III, where we explore four combinations of LLMs for both internal and external feature extraction tasks. It can be seen that using a more powerful LLM (e.g., LLama2-13B) outperforms the combination of using a smaller model in one of the tasks, and improves both AUC (area under the curve) and ACC (accuracy) metrics. In particular, the model reaches its best performance when both the internal and external tasks are performed using LLama2-13B, with an AUC of 82.4 and an ACC of 74.9. This is a significant improvement compared to the LLama2-7B combination.

These experimental results suggest that the performance improvement is mainly due to the fact that when using the more powerful LLama2-13B model for both internal and external feature extraction, the model is able to better capture subtle inconsistencies in the generated responses, thereby leading to

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TABLE III
PERFORMANCE EVALUATION OF CROSS-MODEL COMBINATIONS ON NQ

Internal Feature	External Feature	AUC	ACC
LLama2-7b	LLama2-7b	81.4	74.4
LLama2-7b	LLama2-13b	81.9	74.8
LLama2-13b	LLama2-7b	81.3	74.2
LLama2-13b	LLama2-13b	82.4	74.9

TABLE IV
EVALUATING PROBE CONSTRUCTION WITH INTERNAL REPRESENTATIONS FROM DIFFERENT LAYERS

Layer	Token	AUC	ACC
first-layer	last token	53.6	54.4
middle-layer	last token	81.4	74.4
last-layer	last token	75.3	70.1
middle-layer	average token	77.6	71.9

improved hallucination detection. In addition, the combination of LLama2-7B for internal feature extraction and LLama2-13B for external feature extraction performs better than the opposite combination (i.e., LLama2-13B for internal and LLama2-7B for external), suggesting that it is more beneficial to use a more powerful model in the external feature extraction stage. The results of this ablation experiment highlight the importance of choosing the right model for different task phases and provide an important reference for further researching on how to fine-tune the choice of LLM.

G. Ablation Studies on Probe Construction

In this section, we investigate the effect of feature selection for LLM on the experimental results. We select the hidden layer states of the first, middle (16th), and last layers of the LLM for our experiments, and further, we investigate the impact of selecting either the average token or the last token of a particular hidden layer as its hidden state on the experimental outcomes. The results are shown in the Table IV, showing that the best results are achieved when the last token in the middle layer is chosen as the hidden layer state and used as the default setting for the experiments.

H. Ablation Studies on Hyperparameter k

Fig. 3 illustrates the effect of the number of generated responses k on classification performance in terms of accuracy (ACC) and area under the curve (AUC) on the NQ dataset. As shown in the figure, both ACC and AUC generally increase as increasing k, though AUC exhibits a more pronounced upward trend. Specifically, AUC increases from 68.2 at $k = 3$ to 74.8 at $k = 11$, indicating that aggregating more diverse responses enhances the model's ability to distinguish factual from hallucinatory outputs. By contrast, ACC remains relatively stable, with small fluctuations and a peak of 81.4 at $k = 9$, suggesting

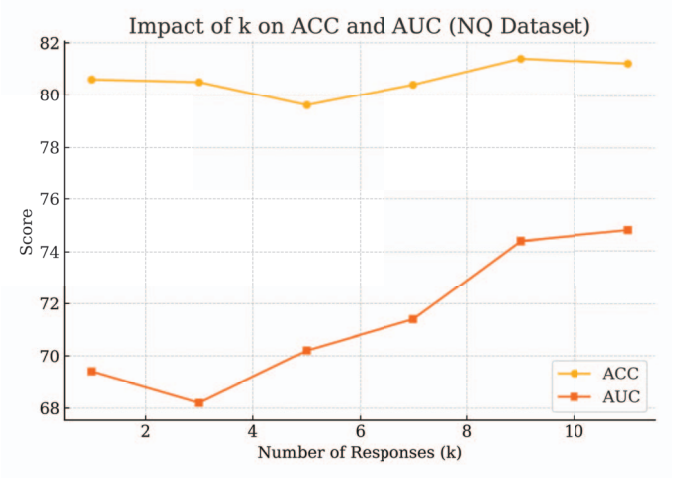


Fig. 3. Impact of the number of generated responses k on performance (NQ dataset).

that classification accuracy saturates earlier than AUC. These observations support the hypothesis that leveraging semantic diversity through multi-response generation strengthens factuality detection. The external features derived from response consistency become increasingly informative as k grows, leading to more reliable and robust model predictions.

VI. CONCLUSION

In this paper, we proposed **ESIF**, a novel hallucination detection framework for large language models (LLMs), which effectively integrates internal hidden states and external semantic distribution features. By leveraging internal layer representations, ESIF captures the model's latent belief structure and confidence signals, while the use of semantic consistency across multiple generated responses provides an external view of factual alignment. This dual-perspective design allows ESIF to detect hallucinations with both high accuracy and strong transferability.

Extensive experiments on three benchmark datasets—NQ, TriviaQ, and WebQ—demonstrate that ESIF consistently outperforms existing baselines, including both probing-based and consistency-based approaches. In particular, our ablation analysis reveals that internal features contribute to classification accuracy, especially in scenarios where LLMs confidently generate incorrect outputs, while external semantic features significantly improve robustness across domains with distributional shifts.

Moreover, we analyzed the impact of varying the number of generated responses k on model performance. Results show that AUC improves steadily with larger k, highlighting the value of sampling semantic variations to capture uncertainty and enhance detection reliability. The incorporation of an attention mechanism further enables the model to dynamically balance the contribution of internal and external signals, leading to improved context-awareness and performance.

Overall, ESIF offers a principled, modular, and interpretable solution for hallucination detection in LLMs, with strong

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generalization ability and potential for application in real-world question answering and generation scenarios. Future work includes extending ESIF to multi-modal settings and integrating symbolic

VII. Limitations

Our work is limited by the following: (1) Manual annotation: during data preparation, we need to manually go through the process of calibrating with the reference answers to obtain the phantom labels for LLM-generated responses , which requires a large amount of time and cost for a large-scale training set. (2) Open-source LLM: Since our approach needs to make use of the internal states of the LLM, such as the hidden layer states, the LLM we use needs to be open-source to obtain the relevant data. (3) Factual errors: ESIF is limited to detect factual error detection of responses in a QA task, and is unable to detect other errors such as grammatical errors and logical errors.

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